

ENTERPRISE SERVER DEPLOYMENT AND OPERATING SYSTEM INSTALLATION

Installation Guide

ITEP 414 - System Administration and Maintenance
Week 3 Portfolio Project

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1. Project Overview

This project focuses on the deployment and basic configuration of server operating systems in a virtualized environment. The main task was to install Ubuntu Server using Oracle VirtualBox and configure the necessary settings to prepare the server for basic administration.

The Ubuntu Server virtual machine was configured with appropriate hardware resources, networking, storage, a hostname, an administrative user account, and OpenSSH Server. After installation, several commands were performed to verify that the server was functioning properly, including checking the hostname, IP address, internet connectivity, system updates, and SSH service.

The project also included studying the differences between BIOS and UEFI and understanding the Ubuntu boot process. In addition, Windows Server Evaluation was installed in another virtual machine to provide experience with a different enterprise server operating system. Windows Server, Ubuntu Server, and Rocky Linux were then compared based on their major features and typical enterprise uses.

The purpose of this project is to develop basic skills in server deployment, operating system installation, virtualization, system verification, and technical documentation.

2. Objectives

- Install Ubuntu Server in a virtual machine.
- Configure the required server settings during installation.
- Configure the server hostname and administrative user account.
- Configure the server network using DHCP.
- Install and enable OpenSSH Server for remote administration.
- Verify that the Ubuntu Server installation is functioning correctly.
- Update the installed Ubuntu Server packages.
- Understand the differences between BIOS and UEFI.
- Understand the basic Ubuntu/Linux boot process.
- Install Windows Server Evaluation in a separate virtual machine.
- Compare Windows Server, Ubuntu Server, and Rocky Linux.
- Practice documenting server installation and configuration procedures.

3. Software Requirements

Software	Purpose
Oracle VirtualBox	Used to create and manage the virtual machines.
Ubuntu Server LTS ISO	Used to install the Ubuntu Server operating system.
Windows Server 2022 Evaluation ISO	Used for the Windows Server installation activity.

Oracle VirtualBox provided the virtual environment needed to install the server operating systems without modifying the operating system installed on the physical computer.

4. Virtual Machine Specifications

4.1 Ubuntu Server Virtual Machine

Component	Configuration
Virtual Machine Name	Ubuntu-Server-Week03
Operating System	Ubuntu Server LTS
Memory	4 GB / 4096 MB
Processor	2 Virtual Processors
Virtual Disk	40 GB
Network	NAT
Installation Media	Ubuntu Server LTS ISO
Server Hostname	server01
Network Configuration	DHCP
SSH	OpenSSH Server

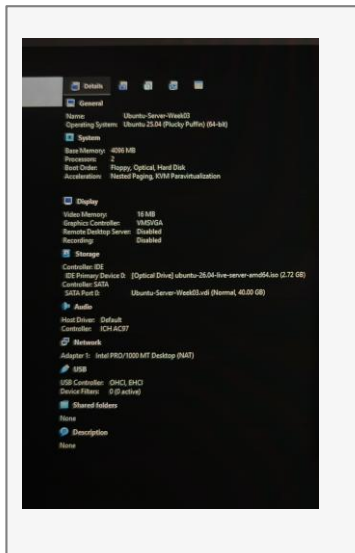


Figure 1. Ubuntu Server Virtual Machine Configuration

4.2 Windows Server Virtual Machine

Component	Configuration
Virtual Machine Name	Windows-Server-Week03
Operating System	Windows Server 2022 Standard Evaluation (Desktop Experience)
Memory	4 GB / 4096 MB
Processor	2 Virtual Processors
Virtual Disk	50 GB
Network	NAT
Installation Media	Windows Server 2022 Evaluation ISO

The Windows Server virtual machine specifications were selected for this implementation. The Week 3 module does not prescribe specific virtual hardware specifications for the Windows Server activity.

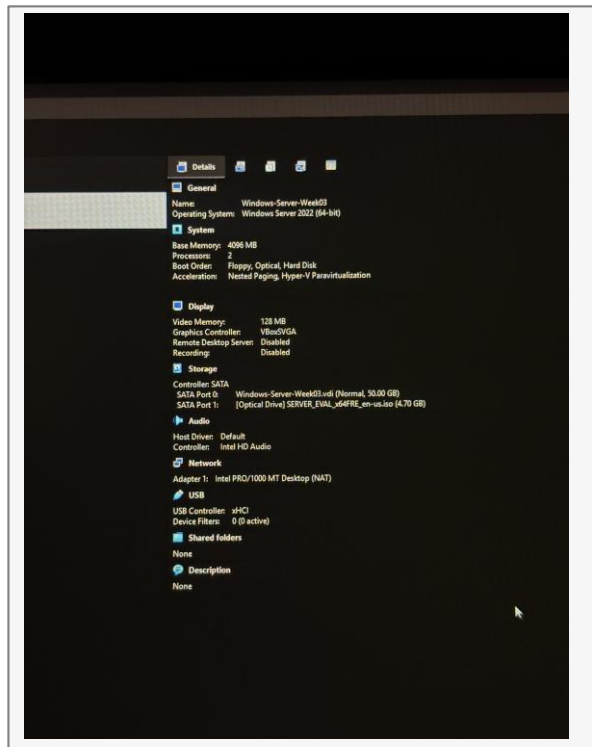


Figure 2. Windows Server Virtual Machine Configuration

5. Ubuntu Server Installation Procedure

5.1 Creating the Virtual Machine

Oracle VirtualBox was opened and a new virtual machine named Ubuntu-Server-Week03 was created. The Ubuntu Server ISO was attached as the installation media. Manual installation was used so that each required configuration could be completed during the Ubuntu Server installation. The virtual machine was allocated 4 GB RAM, 2 virtual processors, a 40 GB virtual disk, and NAT networking.

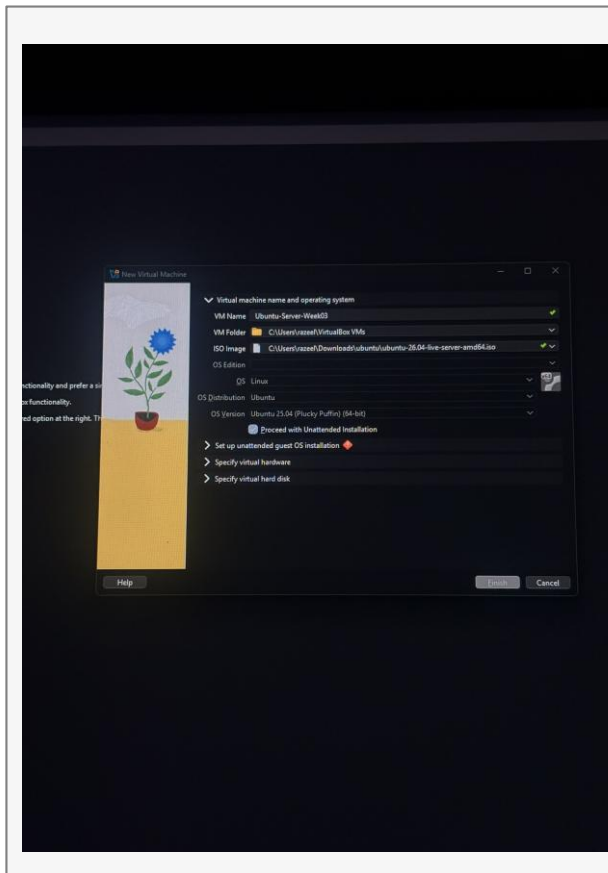


Figure 3. Ubuntu Server VM Setup

5.2 Selecting the Ubuntu Server Installation

After starting the virtual machine, the Ubuntu Server installer was loaded. The normal Ubuntu Server installation option was selected instead of the minimized version. No unnecessary third-party packages were selected.

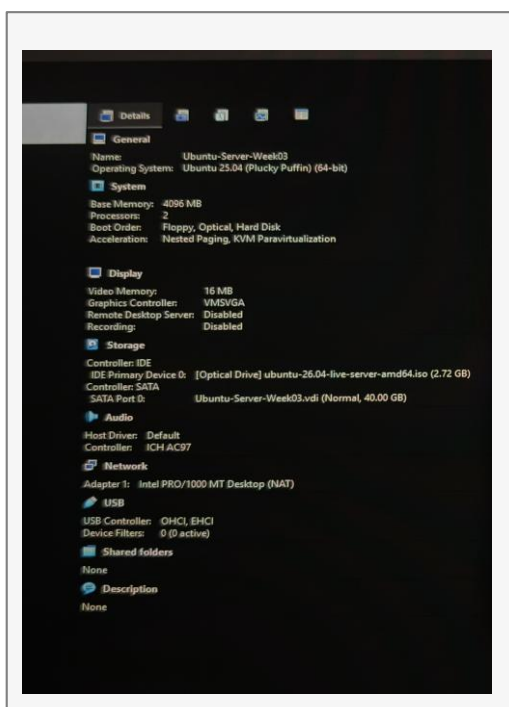


Figure 4. Ubuntu Server Installation Type

5.3 Language Configuration

The installation language was configured as English.

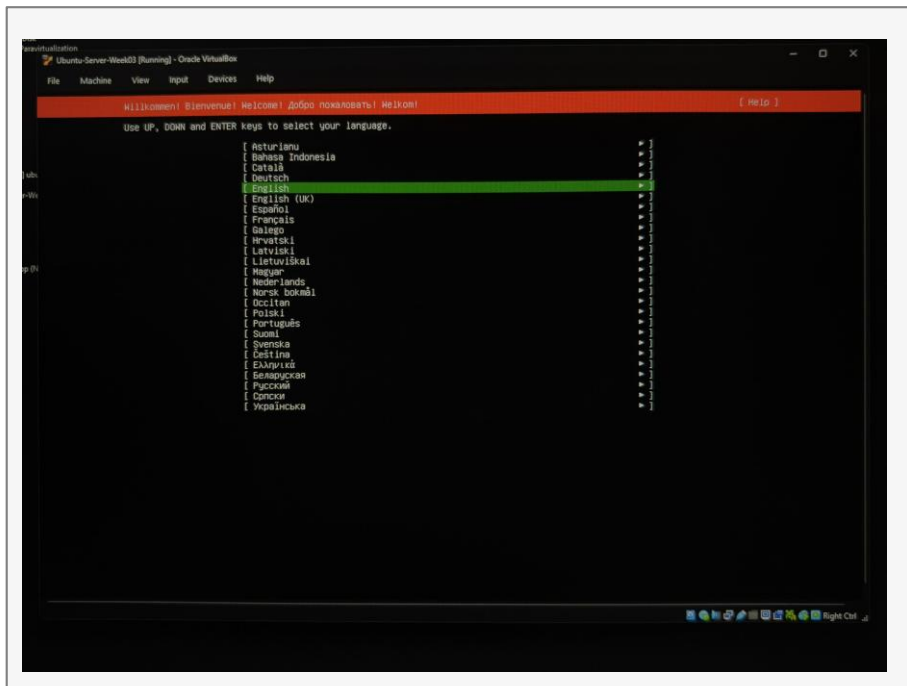


Figure 5. Ubuntu Server Language Configuration

5.4 Keyboard Configuration

The keyboard layout was configured as English (US).

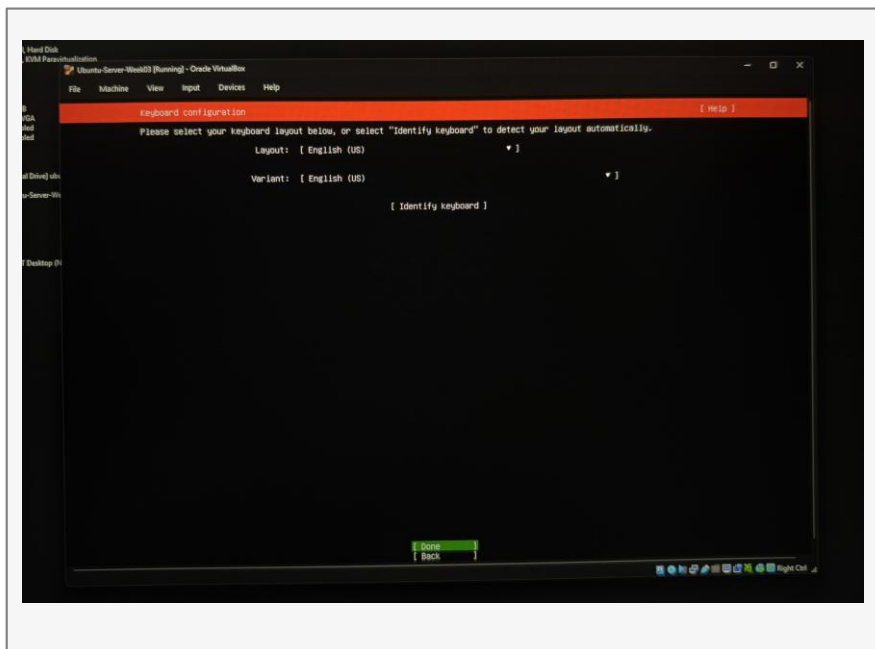


Figure 6. Keyboard Configuration

5.5 Network Configuration

The network was configured using DHCP. The VirtualBox network adapter used NAT. During installation, interface enp0s3 received IPv4 address 10.0.2.15/24. No manual/static IPv4 address was configured.

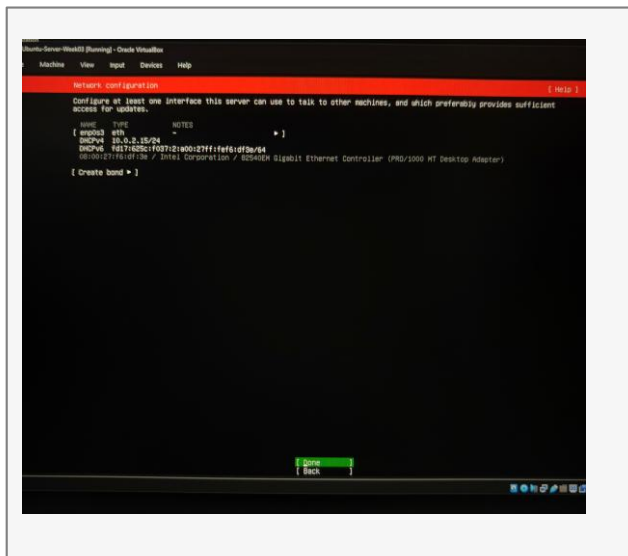


Figure 7. DHCP Network Configuration

5.6 Proxy Configuration

No proxy server was required for the installation. The proxy address field was left blank.

5.7 Ubuntu Archive Mirror

The default Ubuntu archive mirror provided by the installer was retained. No custom mirror address was required.

5.8 Storage Configuration

The storage was configured using Guided - Use Entire Disk. The entire 40 GB VirtualBox virtual disk was selected. LVM (Logical Volume Management) was enabled, and the displayed filesystem included ext4. LUKS encryption was not enabled.

Item	Configuration
Virtual Disk Size	40 GB
Storage Method	Guided / Use Entire Disk
LVM	Enabled
Root Filesystem	ext4
/boot	ext4

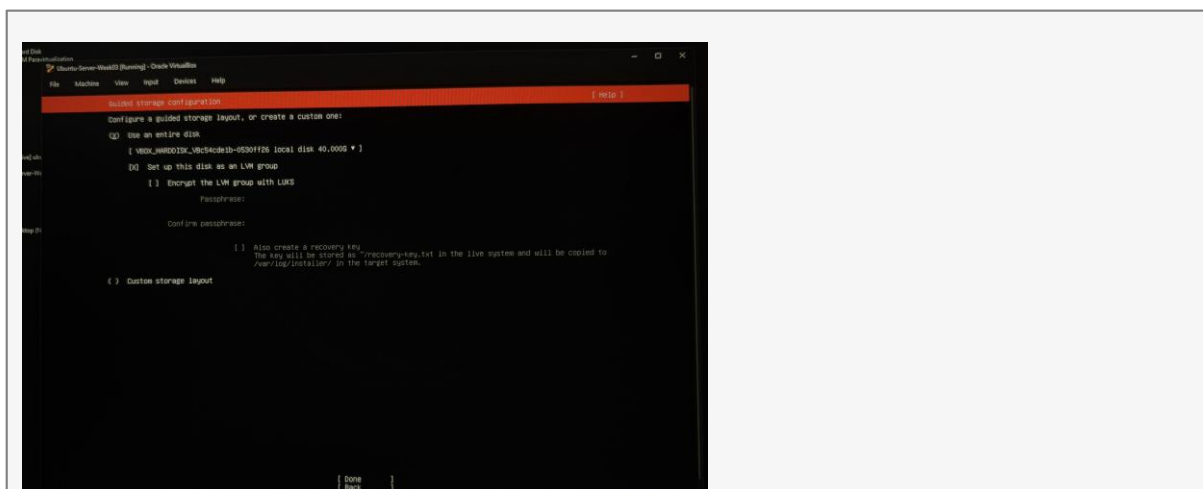


Figure 8. Guided Storage Configuration

6. Configuration Details

6.1 User Account Configuration

A non-root administrative user was created during installation. The account is used for normal server administration, while sudo is used when elevated privileges are required. For security reasons, the account password is not included in this documentation.

6.2 Hostname Configuration

The Ubuntu Server hostname was configured as server01. The VirtualBox VM name is Ubuntu-Server-Week03, while the Ubuntu hostname is server01.

6.3 OpenSSH Server

The Install OpenSSH Server option was enabled during installation. OpenSSH supports secure remote administration using the SSH protocol. No SSH identity was imported.

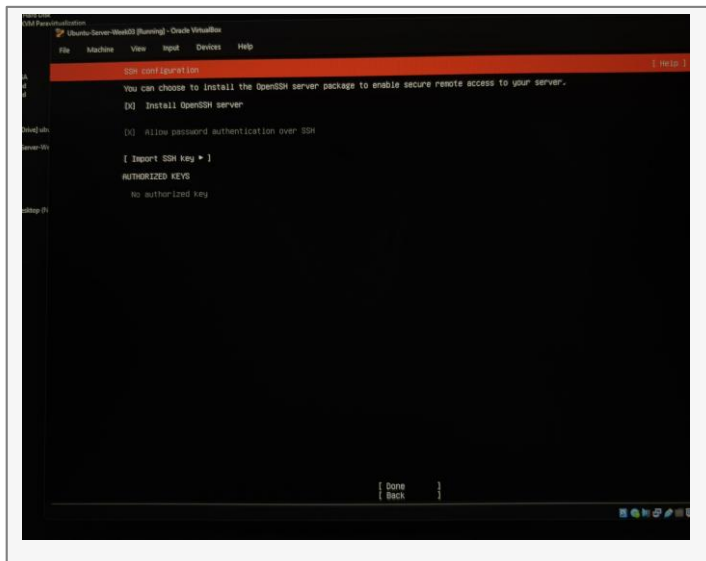


Figure 10. OpenSSH Server Installation

6.4 Additional Packages

No additional packages were selected because they were not required for the activity.

6.5 Completing the Installation

After all required settings were configured, Ubuntu Server installed the operating system and required updates. The virtual machine was restarted, and the installation media was removed when necessary so the VM could boot from the installed system.

7. Server Verification

7.1 Login Verification

The administrative user account created during installation was used to log in to the server. A successful login confirmed that the user account was working properly.



Figure 11. Successful Ubuntu Server Login

7.2 Hostname Verification

The following command was used to verify the hostname:

```
hostname
```

The expected output was server01.

```
razeel@server:~$ hostname
server01
```

Figure 12. Hostname Verification

7.3 IP Address Verification

The following command displays network interfaces and assigned IP addresses:

```
ip addr
```

The server received its IPv4 address through DHCP. During installation, the observed address was 10.0.2.15/24. The final document should use the address shown by the actual verification screenshot if it differs after reboot.

```
razeel@server:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 08:00:27:f6:df:3e brd ff:ff:ff:ff:ff:ff
    altname enx080027f6df3e
    inet 10.0.2.15/24 metric 100 brd 10.0.2.255 scope global dynamic enp0s3
        valid_lft 86104sec preferred_lft 86104sec
    inet6 fd17:625c:f6df:3e::2/64 scope global dynamic mngtmpaddr noprefixroute
        valid_lft 86105sec preferred_lft 14105sec
    inet6 fe80::a00:27ff:f6df:3e/64 scope link proto kernel ll
        valid_lft forever preferred_lft forever
razeel@server:~$
```

7.4 Internet Connectivity Test

```
ping -c 4 google.com
```

Successful replies confirm that the Ubuntu Server can communicate with an external host and has working internet connectivity.

```
razeel@server:~$ ping -c 4 google.com
PING google.com (172.217.26.78) 56(84) bytes of data.
64 bytes from kul08s14-in-f14.1e100.net (172.217.26.78): icmp_seq=1 ttl=64 time=13.1 ms
64 bytes from kul08s14-in-f14.1e100.net (172.217.26.78): icmp_seq=2 ttl=64 time=10.6 ms
64 bytes from kul08s14-in-f14.1e100.net (172.217.26.78): icmp_seq=3 ttl=64 time=9.87 ms
64 bytes from kul08s14-in-f14.1e100.net (172.217.26.78): icmp_seq=4 ttl=64 time=11.6 ms

--- google.com ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3401ms
rtt min/avg/max/mdev = 9.868/11.274/13.052/1.200 ms
razeel@server:~$
```

Figure 14. Internet Connectivity Test

7.5 System Update

```
sudo apt update
sudo apt upgrade -y
```

These commands update package information and install available upgrades.

```
rtt min/avg/max/mdev = 9.868/11.274/13.052/1.200 ms
razeel@server:~$ sudo apt update
[sudo: authenticate] Password:
Hit:1 http://ph.archive.ubuntu.com/ubuntu resolute InRelease
Hit:2 http://ph.archive.ubuntu.com/ubuntu resolute-updates InRelease
Hit:3 http://ph.archive.ubuntu.com/ubuntu resolute-backports InRelease
Hit:4 http://security.ubuntu.com/ubuntu resolute-security InRelease
53 packages can be upgraded. Run 'apt list --upgradable' to see them.
razeel@server:~$
```

Figure 15. Ubuntu Server System Update

7.6 SSH Service Verification

```
systemctl status ssh
```

The expected status is active (running), confirming that OpenSSH Server is installed and running.

```

razeel@server:~$ systemctl status ssh
● ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; disabled; preset: enabled)
   Active: active (running) since Fri 2026-08-14 17:53:46 UTC; 8min ago
     Invocation: 4cd323ded4a94481972c1f800f0e6546
   TriggeredBy: ● ssh.socket
     Docs: man:sshd(8)
           man:sshd_config(5)
    Process: 1175 ExecStartPre=/usr/sbin/sshd -t (code=exited, status=0/SUCCESS)
   Main PID: 1178 (sshd)
      Tasks: 1 (limit: 1700)
     Memory: 1.6M (peak: 2.3M)
        CPU: 26ms
     CGroup: /system.slice/ssh.service
            └─1178 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Aug 14 17:53:46 server systemd[1]: Starting ssh.service - OpenBSD Secure Shell server...
Aug 14 17:53:46 server sshd[1178]: Server listening on 0.0.0.0 port 22.
Aug 14 17:53:46 server sshd[1178]: Server listening on :: port 22.
Aug 14 17:53:46 server systemd[1]: Started ssh.service - OpenBSD Secure Shell server.
razeel@server:~$

```

Figure 16. SSH Service Verification

7.7 Verification Summary

Verification	Command/Method	Expected/Observed Result
Login	User login	Successful
Hostname	hostname	server01
IP Address	ip addr	DHCP address assigned
Internet	ping -c 4 google.com	Successful responses
Package Update	sudo apt update	Package information updated
Package Upgrade	sudo apt upgrade -y	Available upgrades processed
SSH	systemctl status ssh	Active (running)

8. BIOS vs UEFI Comparison

Feature	BIOS	UEFI
Definition	Traditional firmware interface used during system startup.	Modern firmware interface used during system startup.
Boot Process	Traditional BIOS boot mechanism.	Can load EFI boot applications from the EFI System Partition.
Disk Support	Commonly associated with MBR and its limitations.	Commonly supports GPT and larger disks.
Partition Style	MBR	GPT
Security	Limited modern boot security capabilities.	Supports features such as Secure Boot.

Speed	Older boot architecture.	Designed for modern and more flexible boot environments.
Advantages	Simple and compatible with older systems.	Modern features, GPT support, Secure Boot and improved flexibility.
Disadvantages	Older design with storage and feature limitations.	Older hardware/software may have compatibility limitations.
Feature	BIOS	UEFI
Modern Usage	Mainly legacy systems.	Common standard on modern computers.

Why UEFI Has Largely Replaced BIOS

UEFI, or Unified Extensible Firmware Interface, has largely replaced BIOS because it provides features that are more appropriate for modern computers. Both BIOS and UEFI initialize hardware and help start the operating system, but BIOS is an older technology with limitations that are less suitable for newer hardware.

One important advantage of UEFI is its support for modern storage configurations. Traditional BIOS is commonly associated with MBR partitioning, while UEFI commonly works with GPT. GPT supports larger storage devices and more partitions, making it more suitable for modern computers and servers.

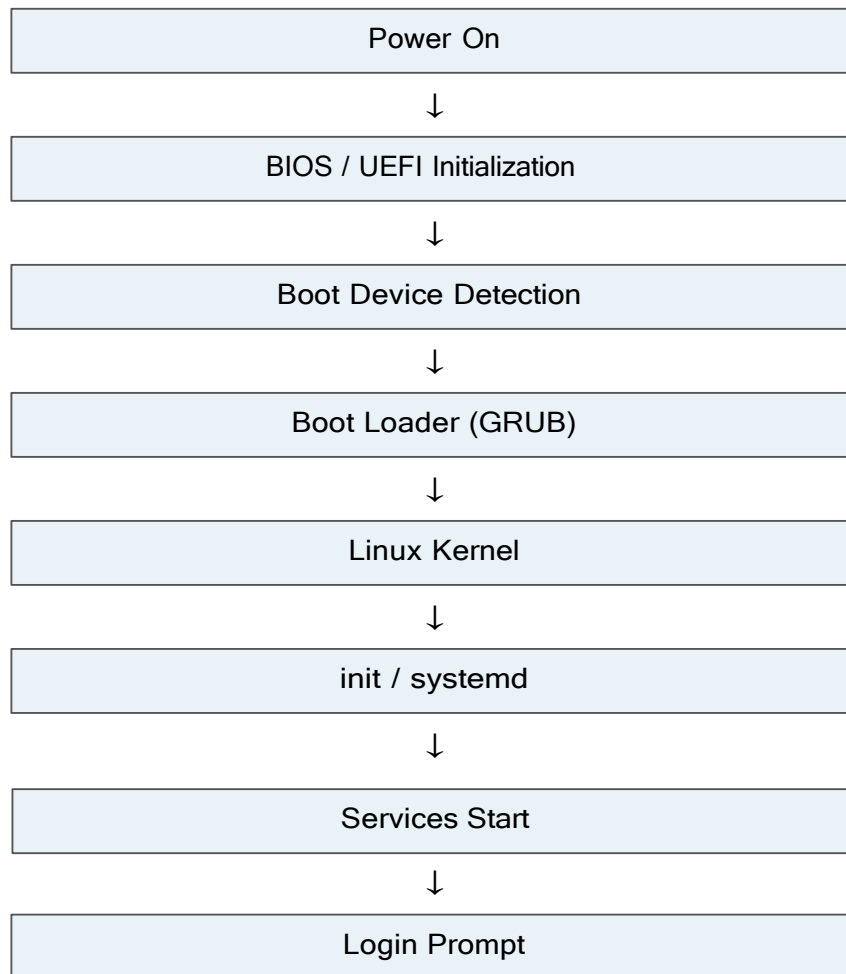
UEFI also provides a more modern boot environment. Instead of depending only on the traditional BIOS boot process, UEFI can work with boot files stored in an EFI System Partition. This provides greater flexibility for modern operating systems.

Security is another important advantage. UEFI can support Secure Boot, which helps prevent unauthorized boot software from loading during startup. This adds protection during the early stages of the computer's boot process.

BIOS remains important because it was widely used for many years and continues to exist in older systems. However, its older design has limitations compared with modern firmware requirements.

Overall, UEFI has largely replaced BIOS because it provides better support for modern storage, newer operating systems, improved boot capabilities, and additional security features. Understanding both technologies is useful for system administrators because firmware configuration directly affects how computers and servers start their operating systems.

9. Ubuntu Boot Process



Power On: The computer or virtual machine receives power and begins startup.

BIOS/UEFI Initialization: The system firmware initializes the hardware and prepares the system for booting.

Boot Device Detection: The firmware identifies a bootable device containing the operating system.

Boot Loader (GRUB): GRUB loads and begins the process of loading Linux.

Linux Kernel: The Linux kernel is loaded into memory and begins managing hardware and system resources.

init/systemd: The system initialization process begins. Modern Ubuntu systems use systemd.

Services Start: Required system services and background processes are started.

Login Prompt: Ubuntu Server displays the login prompt and is ready for administration.

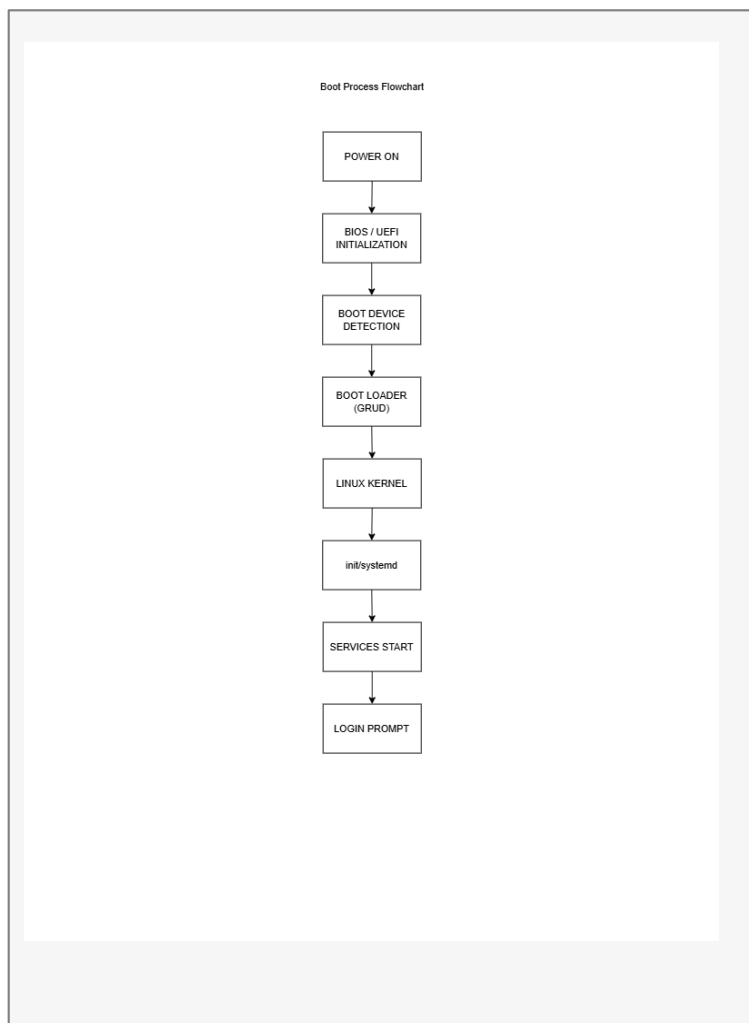


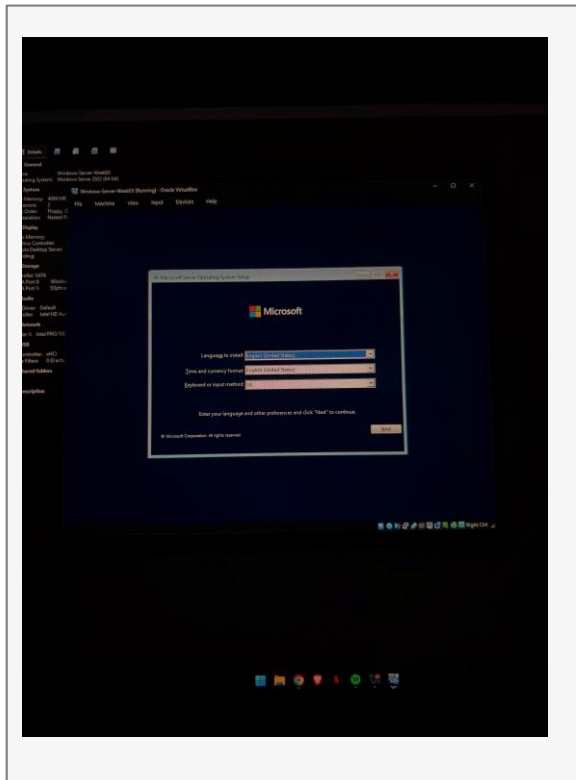
Figure 17. Ubuntu Server Boot Process Flowchart

10. Windows Server Installation Summary

As part of the bring-home activity, Windows Server Evaluation was installed in a separate Oracle VirtualBox virtual machine named Windows-Server-Week03. For this implementation, Windows Server 2022 Standard Evaluation (Desktop Experience) was selected.

10.1 Creating the Virtual Machine

A new VirtualBox virtual machine was created with 4 GB RAM, 2 virtual processors, a 50 GB virtual disk, NAT networking, and the Windows Server 2022 Evaluation ISO.



Windows Server VirtualBox Setup

10.2 Mounting the Windows Server ISO

The Windows Server Evaluation ISO was attached to the virtual optical drive. During the initial attempt, the VM displayed No bootable medium found because the ISO was not mounted correctly. The issue was resolved by attaching the correct ISO.

10.3 Selecting the Windows Server Edition

The selected edition was Windows Server 2022 Standard Evaluation (Desktop Experience).

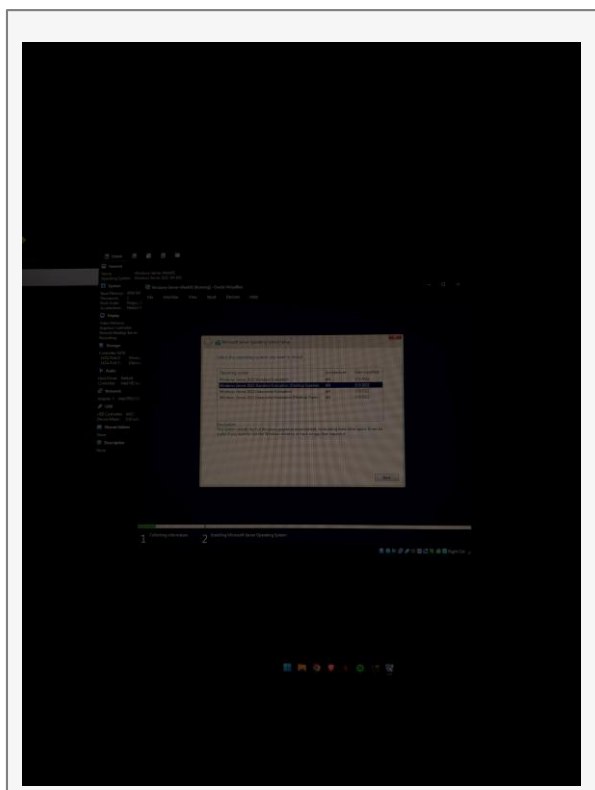


Figure 18. Windows Server Edition Selection

10.4 Installing Windows Server

A clean installation was performed on the virtual hard disk. Windows copied the required files, installed the operating system, and restarted the virtual machine. After installation, the ISO was removed from the virtual optical drive so the VM would boot from the installed operating system.

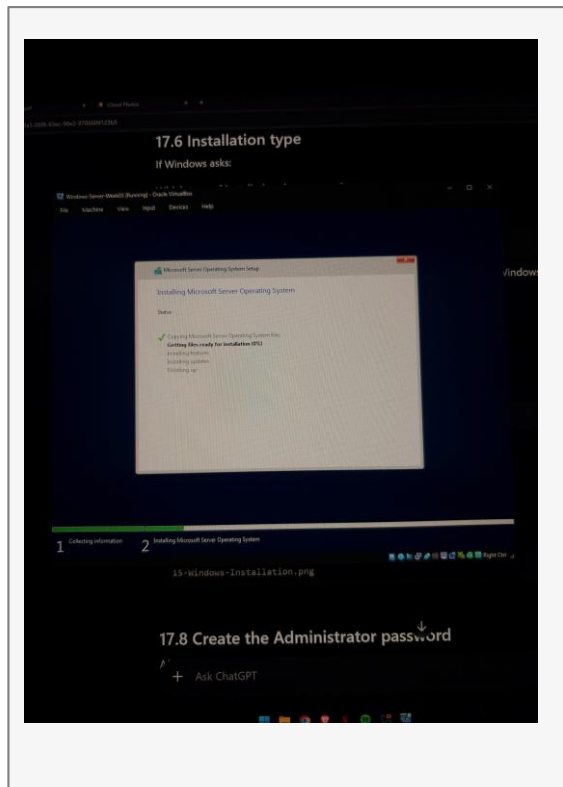


Figure 19. Windows Server Installation

10.5 Administrator Account

A password was created for the built-in Administrator account. The password is not included in this documentation for security purposes. The Administrator account was then used to log in successfully.

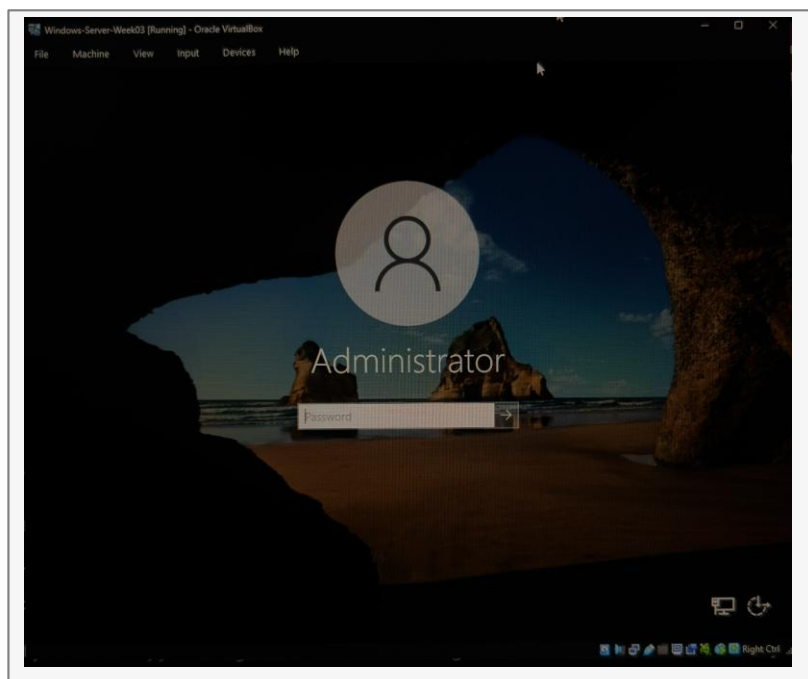


Figure 20. Windows Server Successful Login

10.6 Computer Name

The Windows Server computer name should be recorded here using the actual name configured in the virtual machine: [ENTER YOUR ACTUAL COMPUTER NAME].

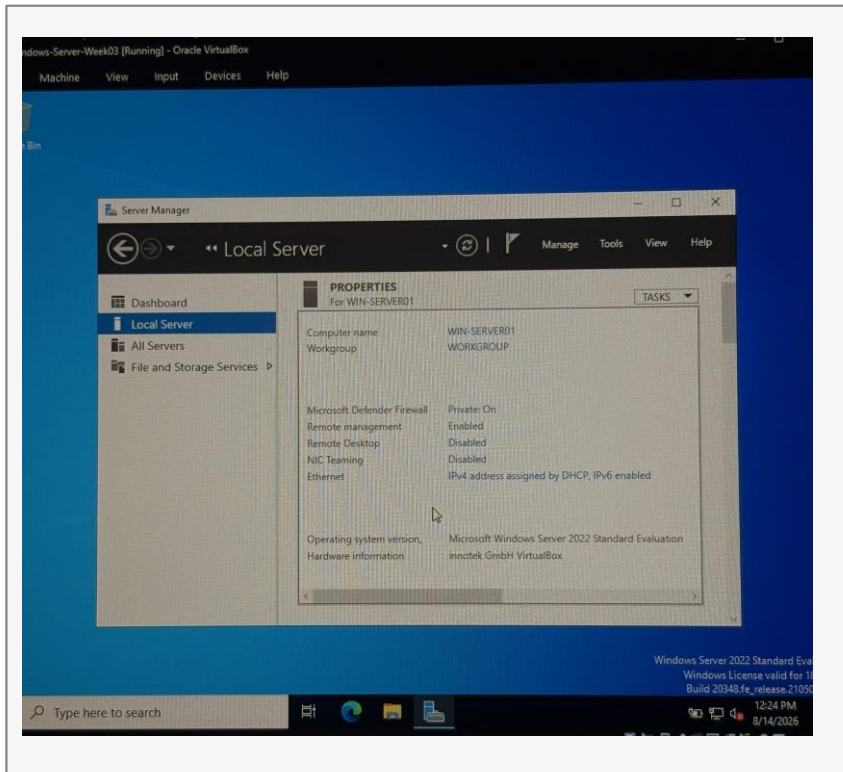


Figure 21. Windows Server Computer Name

11. Operating System Comparison

Category	Windows Server	Ubuntu Server	Rocky Linux
Licensing	Commercial Microsoft server OS; evaluation editions are available.	Open-source Linux distribution.	Open-source Linux distribution.
User Interface	Supports graphical management with Desktop Experience and command-line tools.	Commonly administered through command-line tools.	Commonly administered through command-line tools.
Package Management	Windows-based software/package management methods.	APT and DEB packages.	DNF and RPM packages.
Security	Windows permissions, firewall, updates, and Microsoft enterprise security features.	Linux permissions, security updates, firewall tools and SSH.	Linux permissions, SELinux, firewall and security updates.
Performance	Depends on hardware, roles and configuration.	Suitable for many headless and server workloads.	Designed for stable enterprise Linux workloads.

Category	Windows Server	Ubuntu Server	Rocky Linux
Typical Enterprise Uses	Active Directory, Microsoft-oriented infrastructure, file services and Windows applications.	Web hosting, databases, cloud workloads, containers and network services.	Enterprise Linux services, web/database servers and RHEL-compatible environments.
Advantages	Strong Microsoft ecosystem integration and graphical administration options.	Open source, widely used, flexible and supported by a large ecosystem.	Open source and designed for enterprise/RHEL compatibility.
Disadvantages	Licensing costs may apply and some deployments may require more resources.	Linux command-line administration can have a learning curve.	Linux administration can have a learning curve for beginners.

Discussion

Windows Server, Ubuntu Server, and Rocky Linux are operating systems that can provide services in enterprise environments. However, they differ in licensing, administration, package management, security features, and typical use cases.

Windows Server is developed by Microsoft and is commonly used in organizations that depend on Microsoft technologies. One advantage is the availability of graphical administration through Desktop Experience. It is commonly used for Microsoft-based enterprise infrastructure and services.

Ubuntu Server is an open-source Linux server operating system. It is commonly administered through the command line and uses APT for package management. Ubuntu Server can be used for web servers, databases, cloud environments, containers, and other network services.

Rocky Linux is another open-source Linux distribution designed for enterprise environments and compatibility with the Red Hat Enterprise Linux ecosystem. It uses DNF and RPM for package management and is suitable for organizations that want an enterprise-focused Linux environment.

The appropriate operating system depends on the organization's requirements. System administrators should consider the applications being used, licensing, hardware resources, security requirements, staff experience, and required server services before choosing an operating system.

12. Problems Encountered

Problem 1: Ubuntu Login Incorrect

During the first Ubuntu Server login attempt, the system displayed 'Login incorrect'. At first, it appeared that the password was not being entered because Ubuntu did not display characters while the password was being typed.

Problem 2: Temporary Delay During Ubuntu Installation

During installation, the system spent time installing packages and security updates, and a temporary system message appeared while the VM was processing.

Problem 3: Windows Server - No Bootable Medium

When the Windows Server VM was initially started, VirtualBox displayed 'No bootable medium found!' because the installation ISO was not correctly mounted.

Problem 4: Windows Server Returned to Installation

After Windows Server had been installed and restarted, the system returned to the installer because the Windows Server ISO remained attached to the virtual optical drive.

Problem 5: Ctrl+Alt+Delete Went to the Host Computer

At the Windows Server lock screen, pressing Ctrl+Alt+Delete affected the physical Windows computer instead of the Windows Server virtual machine.

13. Solutions Implemented

Solution 1: Ubuntu Login

The username was checked and the password was carefully entered again. Linux intentionally does not display letters, dots, or asterisks when passwords are typed in the terminal.

Solution 2: Ubuntu Installation Delay

The virtual machine was allowed to continue processing instead of being forcefully shut down.

Solution 3: Mounting the Windows Server ISO

The Windows Server Evaluation ISO was attached to the VM's optical/DVD drive. After mounting the correct ISO, Windows Server Setup booted successfully.

Solution 4: Removing the Installation ISO

After Windows Server was installed, the ISO was removed from the virtual optical drive so the VM could boot from the installed operating system.

Solution 5: Sending Ctrl+Alt+Delete to the VM

VirtualBox's keyboard command was used to send Ctrl+Alt+Delete directly to the Windows Server virtual machine.

These troubleshooting experiences showed that virtual machine problems can be caused by virtual hardware, installation media, boot configuration, or input handling, not only by the operating system.

14. Lessons Learned

This project gave me practical experience in installing and configuring server operating systems using virtualization. By completing the Ubuntu Server installation step by step, I learned how RAM, processors, virtual storage, networking, user accounts, hostnames, and server services work together when deploying a new server.

I also learned the importance of verifying a server after installation. Successfully installing an operating system does not automatically mean that everything is working correctly. Commands such as `hostname`, `ip addr`, `ping`, and `systemctl status ssh` can be used to verify important parts of the system. Updating the server using APT also showed me why system maintenance is necessary after installation.

Another important lesson was understanding the role of BIOS/UEFI and the boot process. I learned that several stages occur before the Ubuntu login prompt appears, including firmware initialization, boot device detection, GRUB, the Linux kernel, `systemd`, and system services.

Installing Windows Server gave me an opportunity to compare Windows and Linux server environments. Windows Server provided a graphical interface through Desktop Experience, while Ubuntu Server relied more heavily on command-line administration.

Finally, I learned that troubleshooting is an important part of system administration. Problems such as an incorrectly mounted ISO, booting from installation media again, or entering passwords in a Linux terminal required me to identify the cause before applying a solution. Overall, this project improved

my understanding of virtualization, server deployment, operating systems, networking, server verification, and basic troubleshooting.

15. References

- Canonical. Ubuntu Server Documentation. <https://documentation.ubuntu.com/server/>
- Oracle. Oracle VirtualBox User Manual. <https://docs.oracle.com/en/virtualization/virtualbox/>
- Microsoft. Windows Server Documentation - Microsoft Learn.
<https://learn.microsoft.com/windows-server/>
- Rocky Linux. Rocky Linux Documentation. <https://docs.rockylinux.org/>
- UEFI Forum. UEFI Specifications and Resources. <https://uefi.org/specifications>

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